



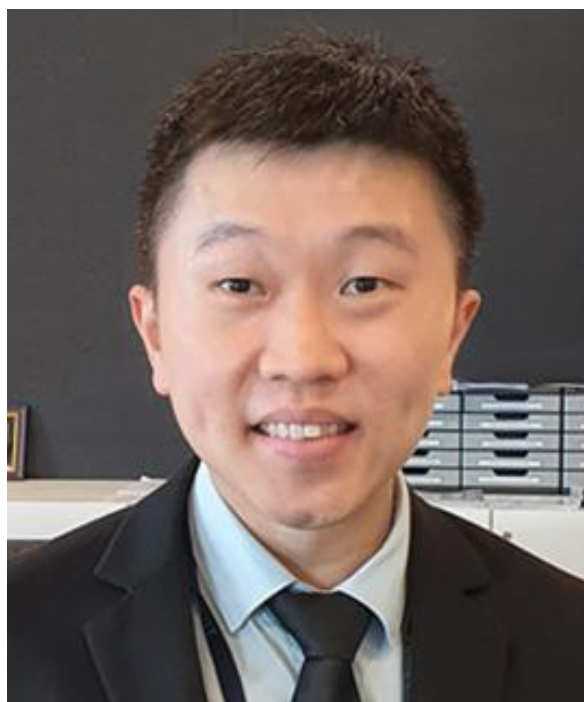
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EEEE3054

Renewable Generation Technologies

Solar Cells



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By end of this module, you will be able to

1. Evaluate different type of energy sources and understand their trends, which may include conventional and non-conventional options depending on the context and application
2. Select appropriate type of renewable energy source and energy storage to mitigate risks associated with security of electrical supply for a given application
3. Analyse various energy storage systems associated with renewable power generation.
4. Determine high level system sizing and component understanding for the renewable power generation.
5. Demonstrate an understanding of various renewable energy sources to design and analyse an integrated renewable energy system.
6. Analyse options (i.e. policies, advices or suggestions) aimed at improving efficiency in renewable energy generation and consumption.
7. Develop pragmatic knowledge of renewable energy sources and systems.



Learning Activities

Week	Monday	Thursday
28	Intro to the Module Solar Cells	Solar Cells
29	HARI RAYA HOLIDAY	
30	GCPV Design	GCPV Design
31	GCPV Design (Example)	Safety Standards, Guidance and Regulation
32	Energy Storage	Energy Storage (Presentation)
33	Efficiency Policy	Efficiency Policy
34	Revision	Revision



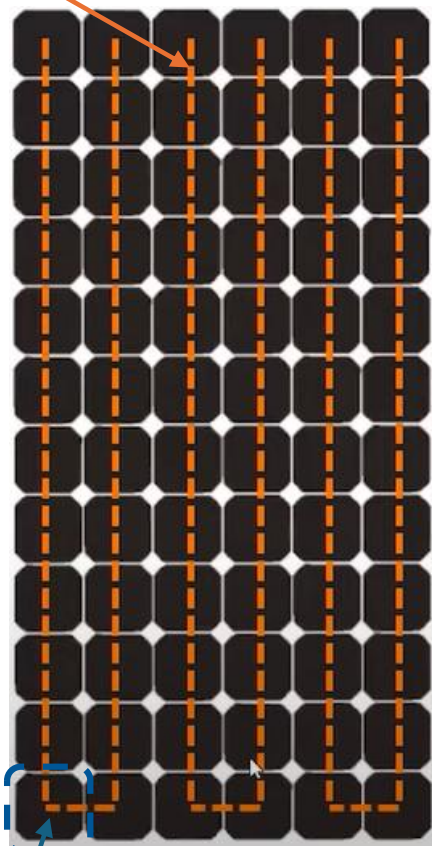
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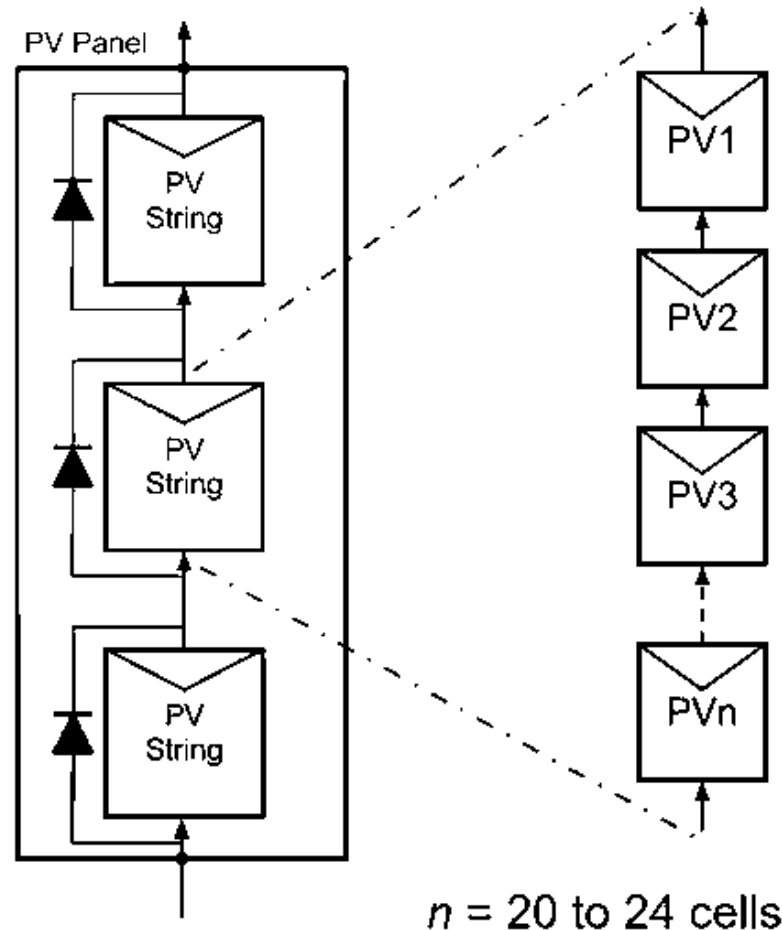
I-V and P-V Curves

Photovoltaic (PV) Module

PV string



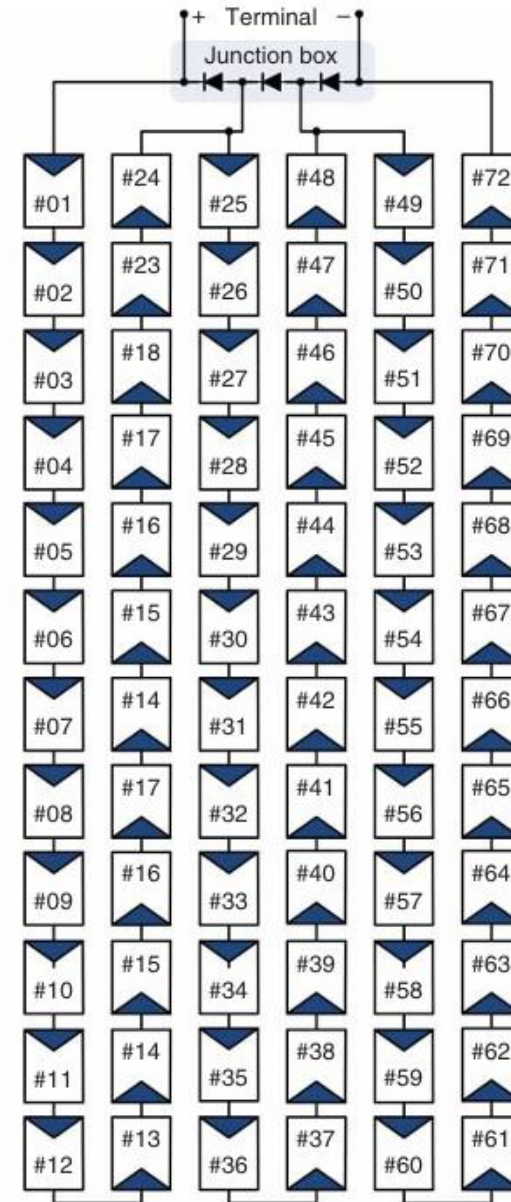
PV cell



- A **PV module** has many cells connected in a string to generate higher voltage.
- The electrical characteristics of each PV cell is different from others due to manufacturing tolerance.

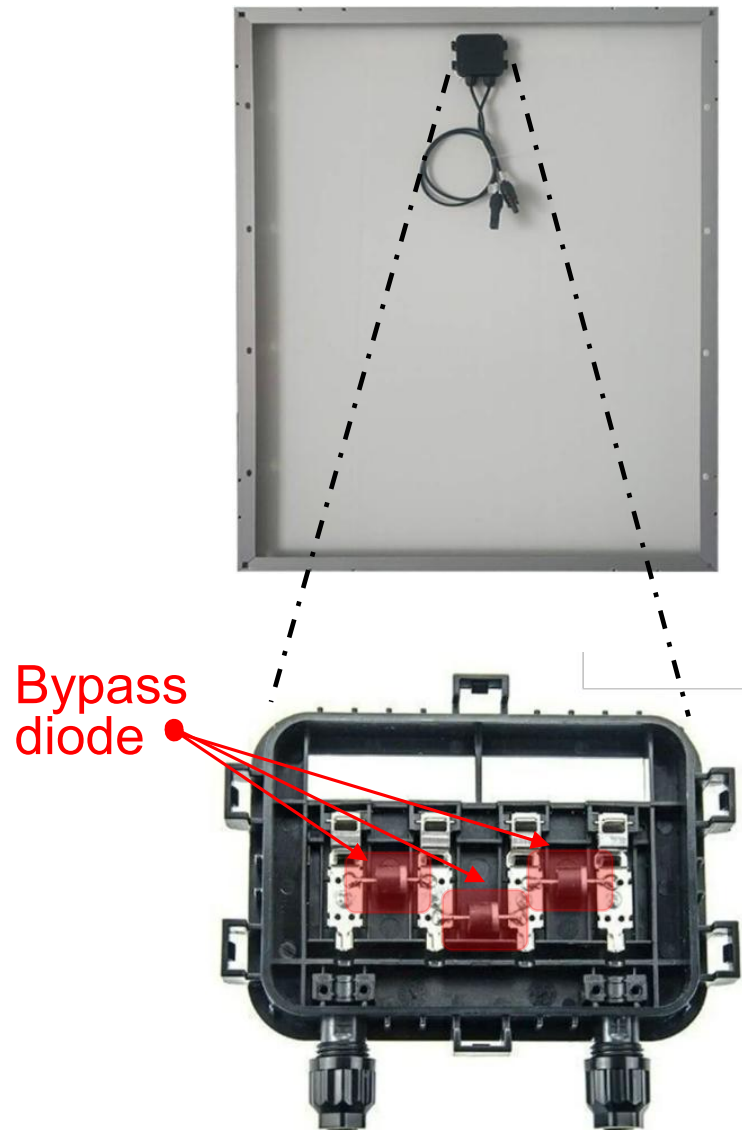


Photovoltaic (PV) Module



72-cell PV module

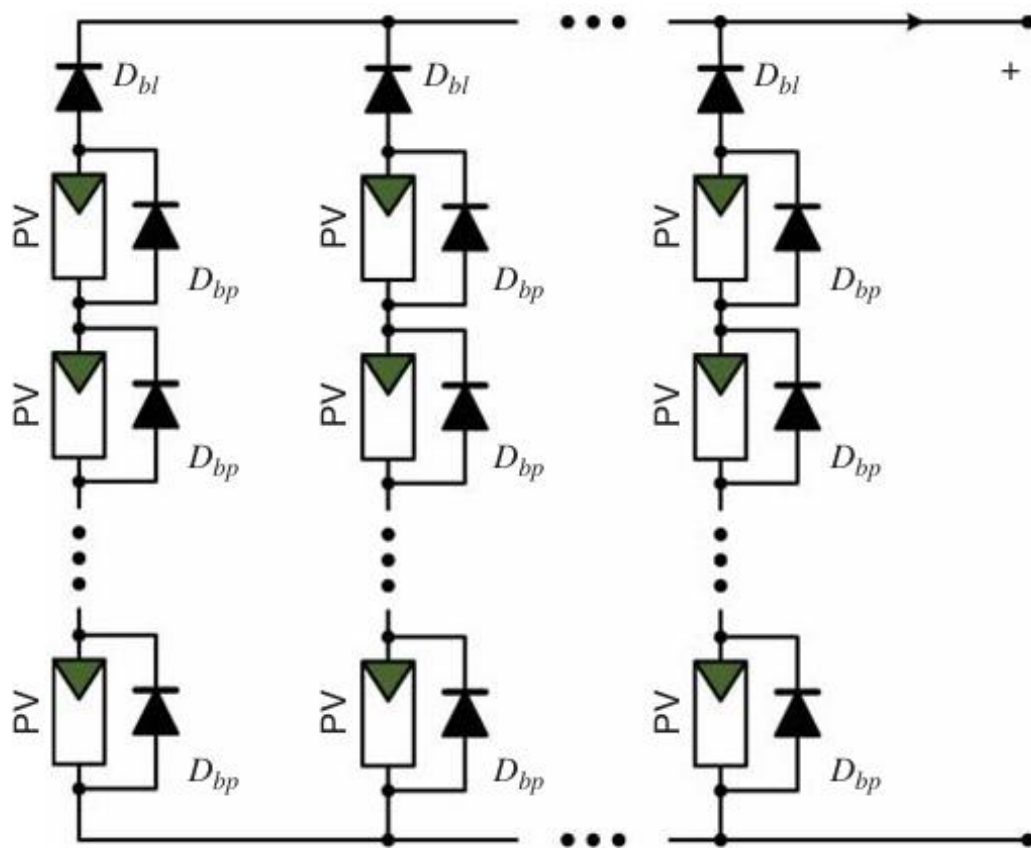
Bypass Diode



- A **bypass diode** is mounted into a junction box on the rear side of the panel and is usually inactive.
- It is connected (in parallel) across PV sub-string (module).

Junction box

Blocking Diode



PV system with bypass and blocking diodes

- During low solar irradiance condition or at night, the PV modules stop producing the power and become idle.
- To avoid the flow of current from load (battery/grid/etc) to PV modules, a diode, called **blocking diode** is used to block the current flow.



Power Mismatch

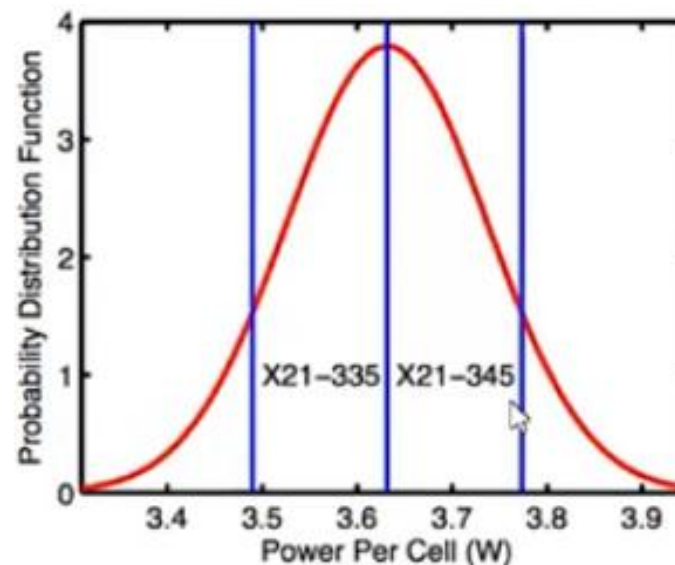
- PV module mismatch, causing significant power losses (often ~10%), occurs when interconnected solar panels operate at different currents or voltages.
- Major causes include
 - ❑ manufacturing tolerance variations
 - ❑ partial shading (leaves, dirt, structures)
 - ❑ uneven aging/degradation of modules
 - ❑ temperature differences
 - ❑ mismatched panel specifications (mixing different wattages)



Mismatch – Manufacturing



Binning is the process of categorising individual PV cells into groups (bins) based on similar electrical performance characteristics to minimise power losses when assembling them into a solar module

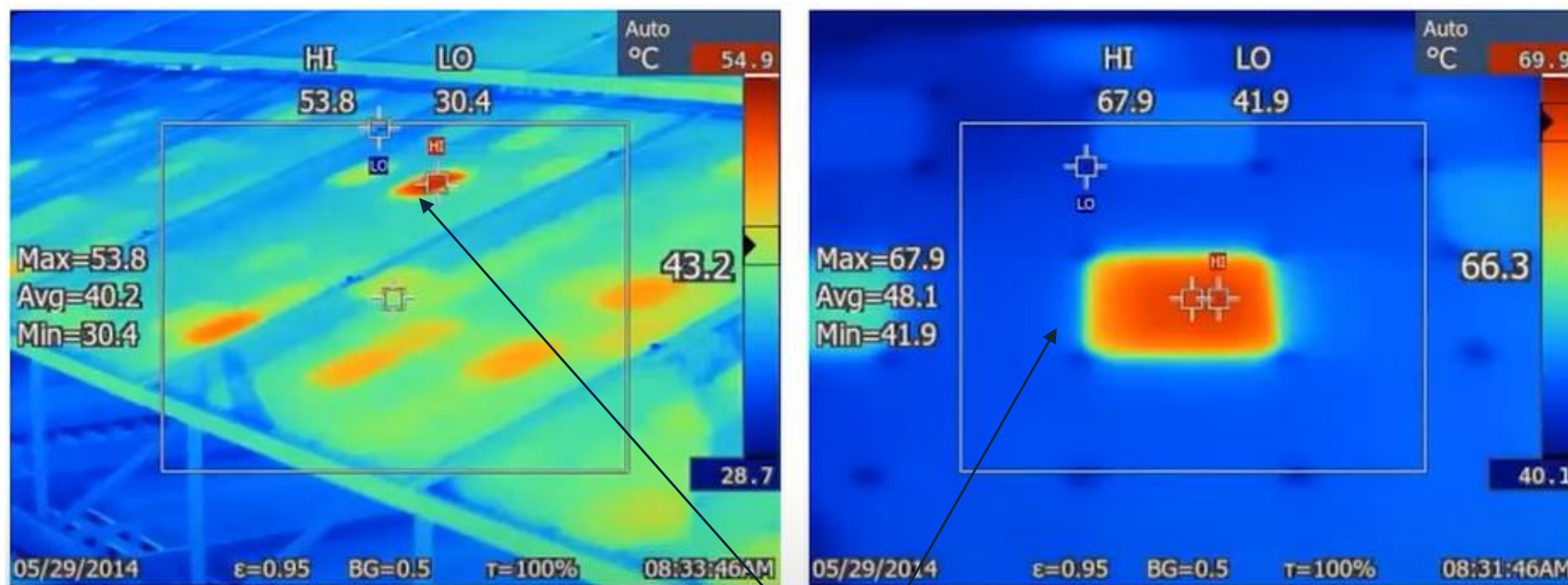


Mismatch – Partial Shading



- Partial shading occurs due to bird drops, shadowing, dust, etc which lead to mismatch losses.

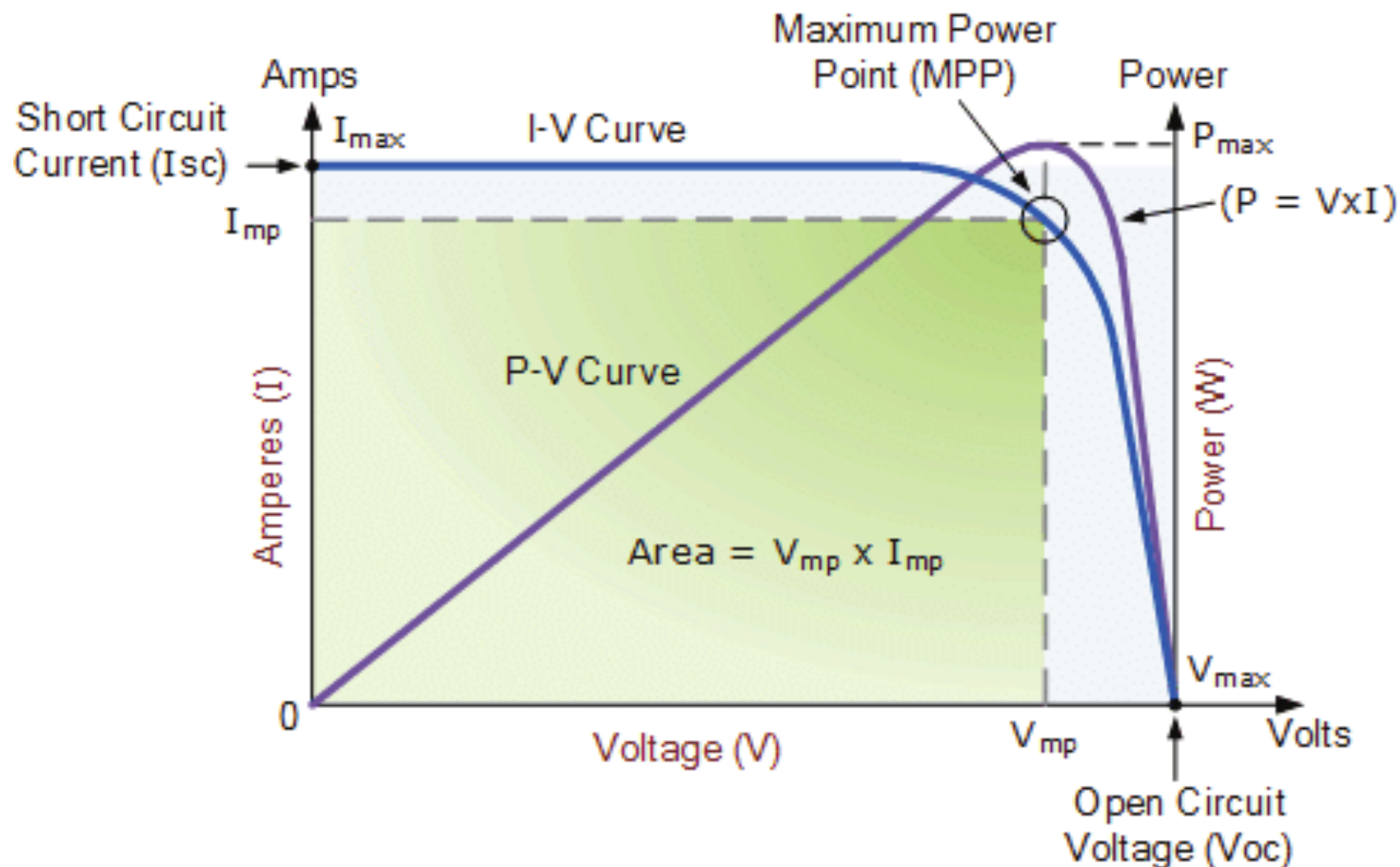
Mismatch – Partial Shading



The power from neighbouring PV cells is absorbed by the shaded cell and it becomes **HOT**.



I-V and P-V Curves





I-V and P-V Curves

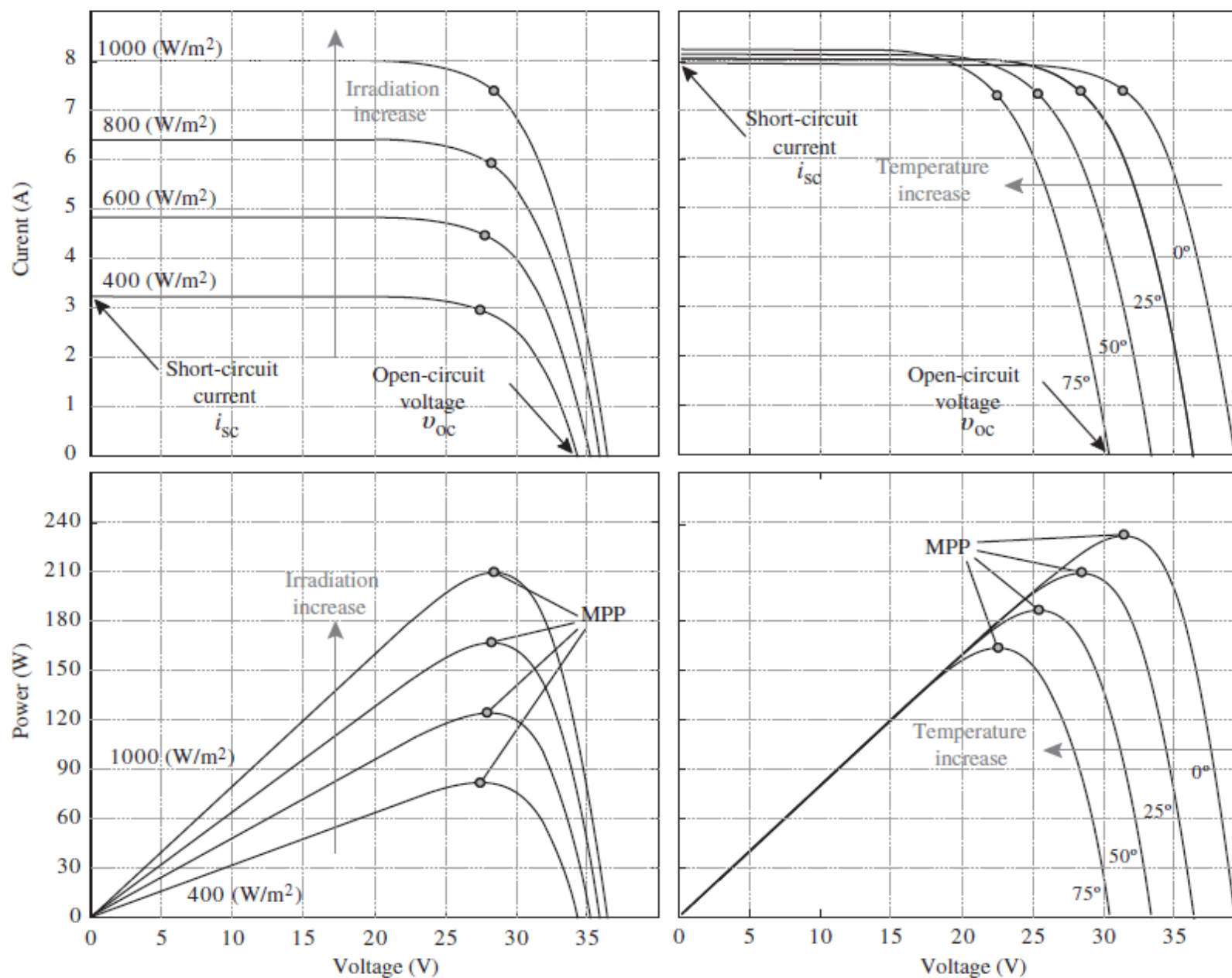
- The temperature has a significant effect on the open-circuit voltage value.
- The short-circuit current is linearly dependent on the irradiance..
- The cell temperature (T), is calculated from the ambient temperature T_a by using the nominal operating cell temperature (NOCT):

$$T = T_a + \frac{NOCT - 20}{800} \cdot G$$

NOCT is defined as the temperature reached by open-circuit cells in a module under the following conditions: irradiance on cell surface $G = 800 \text{ W/m}^2$, air temperature $T_a = 20^\circ\text{C}$, wind velocity = 1 m/s, mounting = open back side

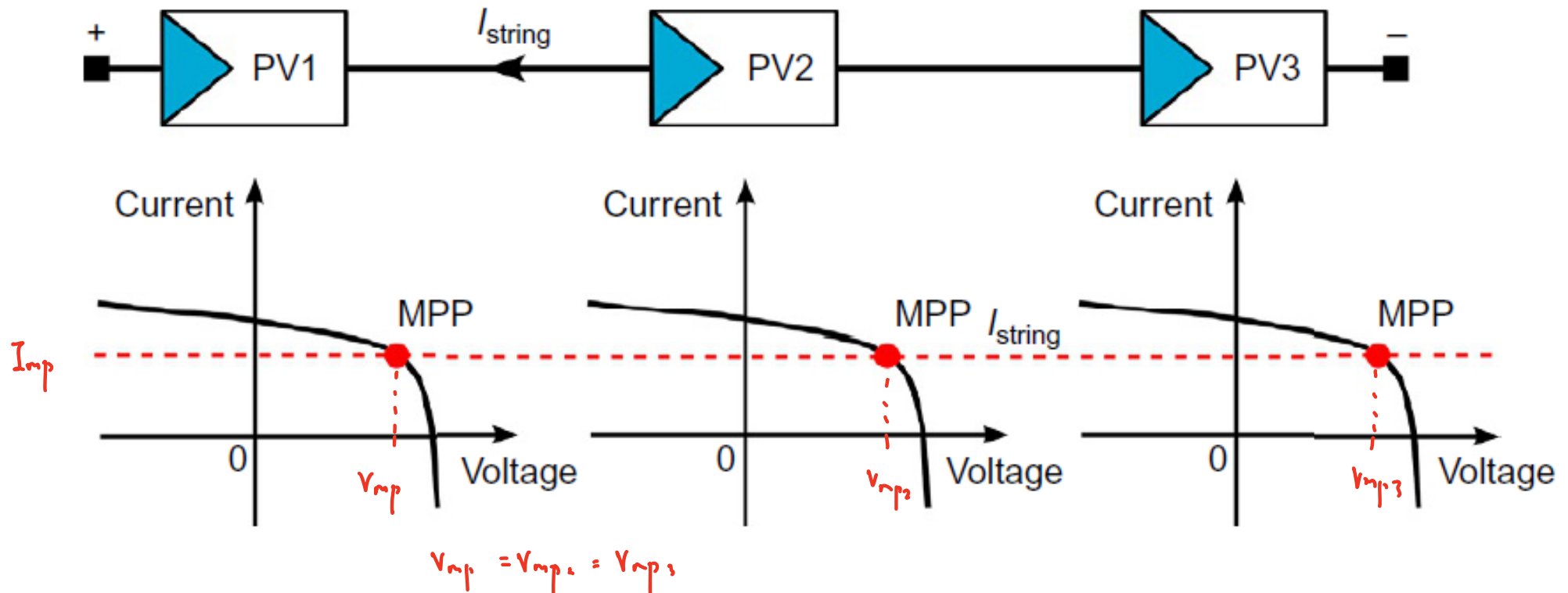


I-V and P-V Curves

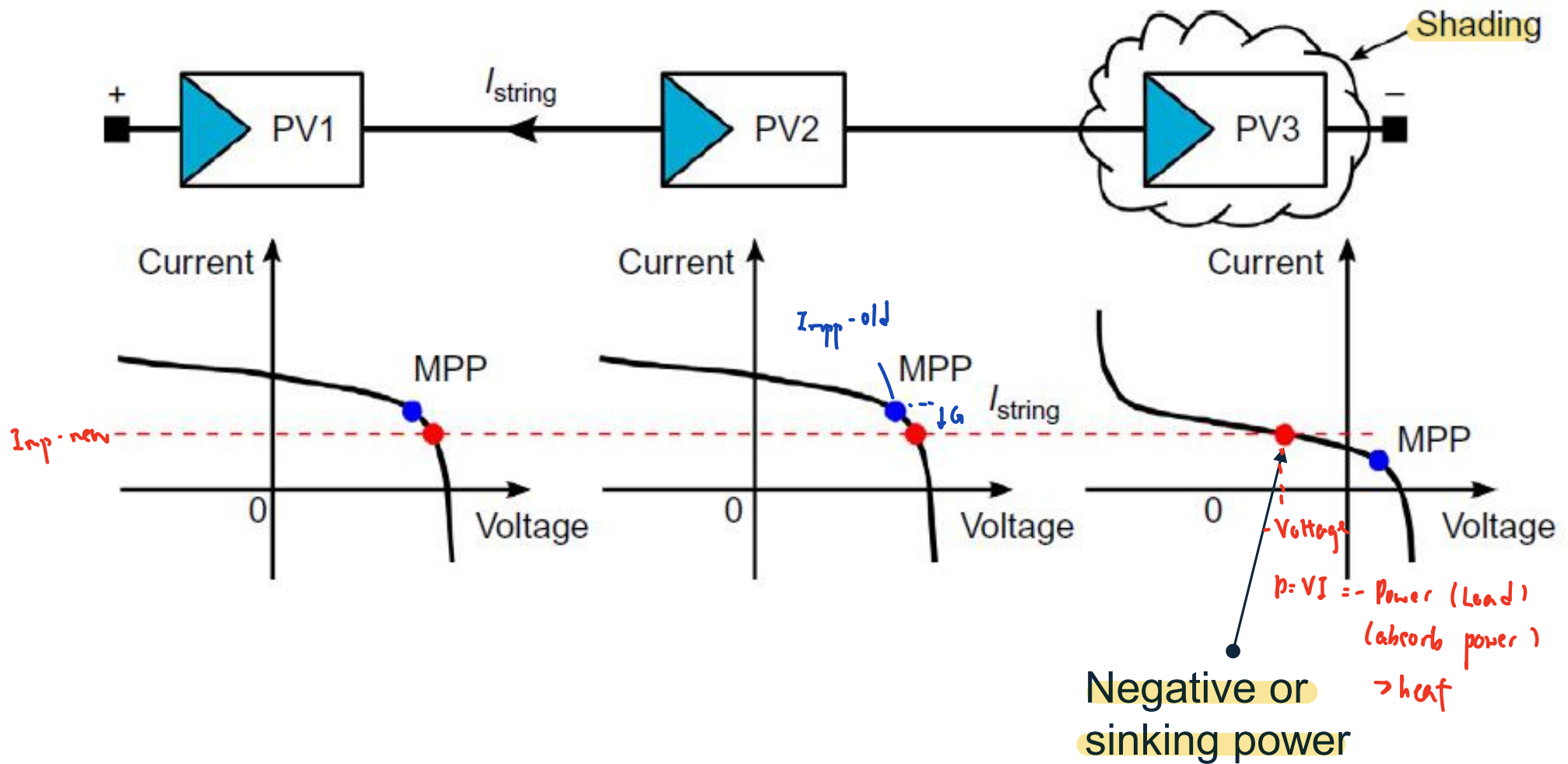




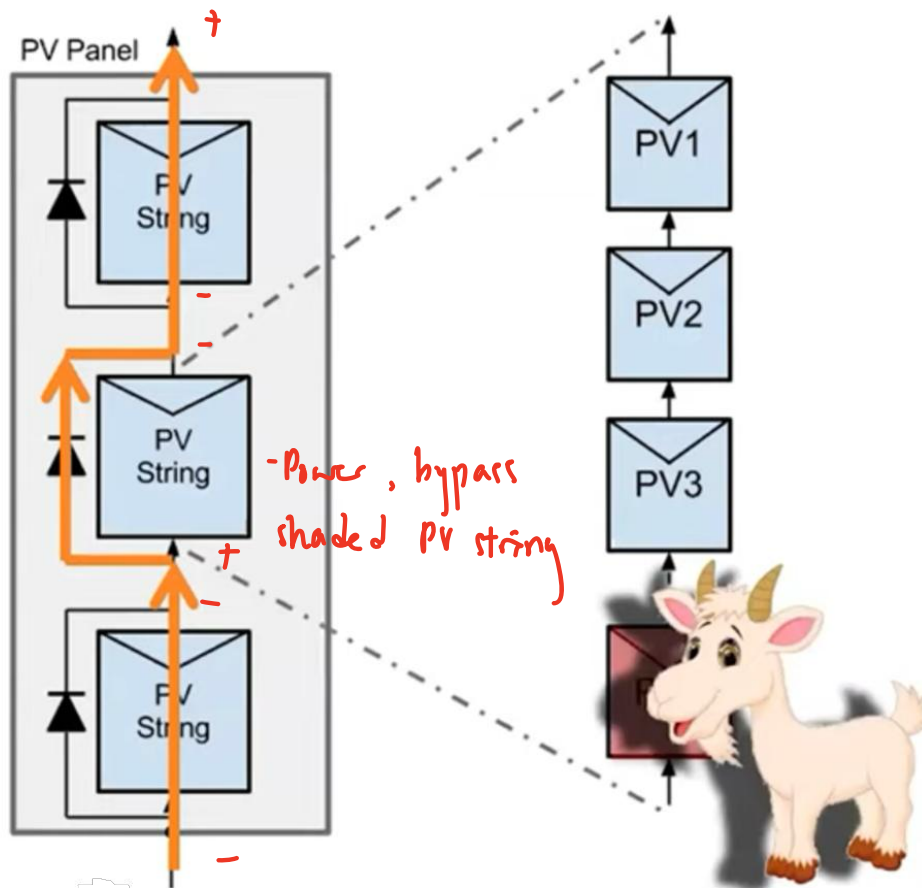
Normal Condition



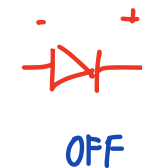
Partial Shading Condition



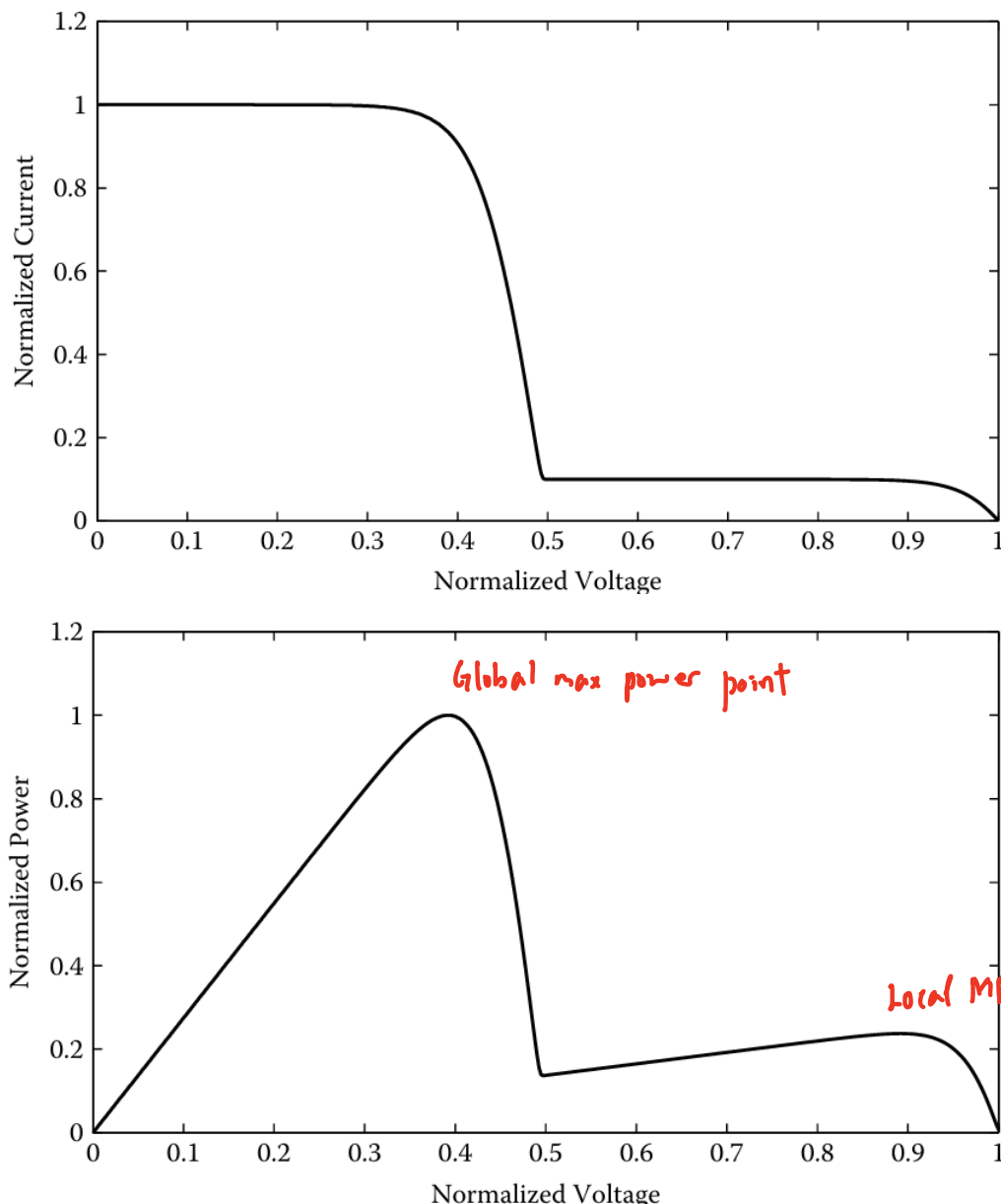
Bypass Diode



- Once that reverse voltage exceeds the diode's forward threshold ($\sim 0.5\text{--}0.7\text{ V}$), the bypass diode conducts.
- The shaded sub-string is bypassed, protecting the cells from damage.



I - V and P - V curves - Partial Shading



- Without bypass diodes, the shaded cells would be forced into reverse bias, causing hot-spots and large power loss.
- With bypass diodes, it bypasses the shaded module when the sub-string voltage becomes negative. That shaded section is bypassed,
- Current can continue to flow through the rest of the module.
- However, this generates multiple peaks in I - V and P - V curves which requires other techniques to harness the maximum power.

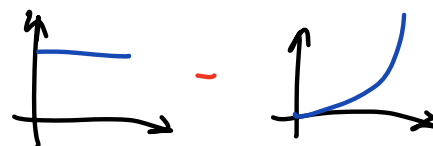


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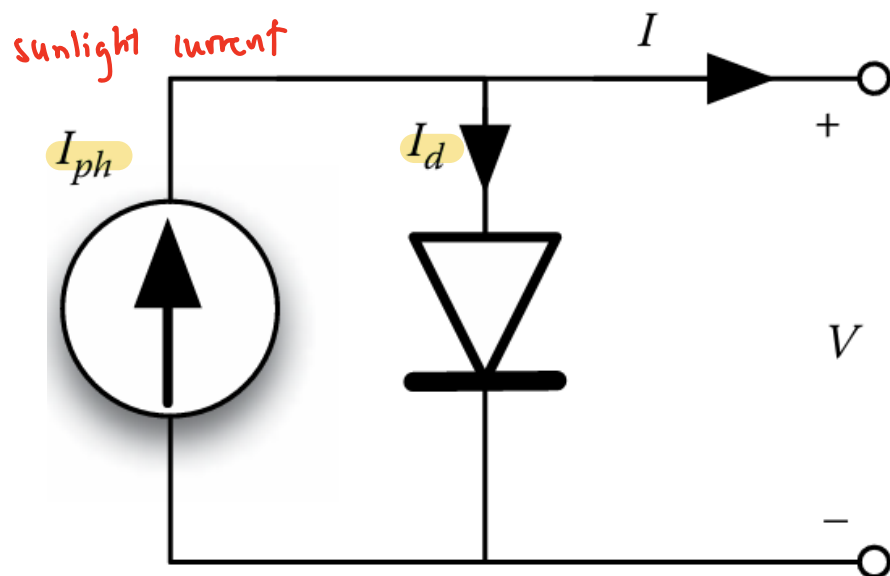
Equivalent Circuit

Ideal Circuit Model



- The I - V curve is similar to a constant current minus diode current, which is nonlinearly dependent on its voltage.
- Therefore, an ideal equivalent circuit is a current source in parallel with a diode.

$$I = I_{ph} - I_d$$

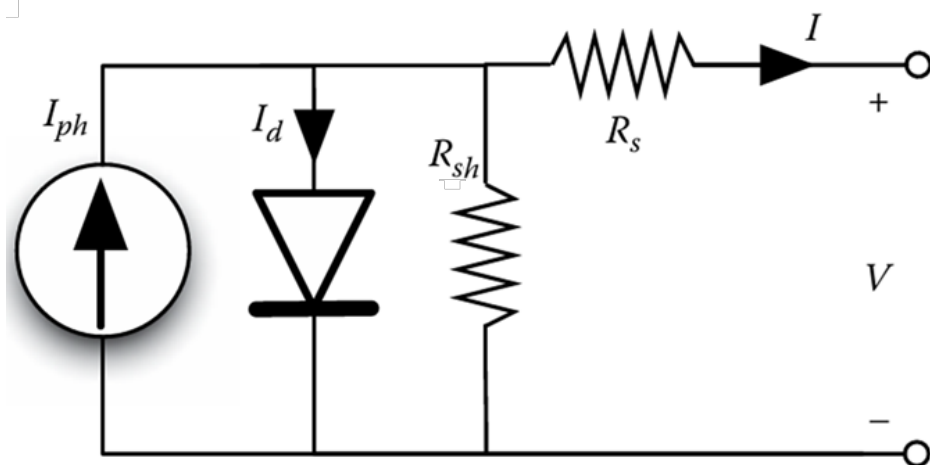


Ideal equivalent circuit of a PV module

- Photocurrent source (I_{ph})** represents the current generated by sunlight (proportional to irradiance).
- Diode (D)** represents the p-n junction of the PV cell. It accounts for the exponential I - V relationship of the semiconductor.

Single Diode Model

- The single-diode model is the most widely used because it balances accuracy with simplicity.
- It represents the PV cell (or module) using standard electrical components that mimic the physical effects inside the solar cell.



Single-diode model

- **Series resistance (R_s)** represents internal resistances (contacts, metal interconnections). Higher R reduces output voltage and power.
- **Shunt resistance (R_{sh})** represents leakage paths across the junction. A high is desirable (minimizes leakage losses).



Single Diode Model

- The output current I of the PV cell is given by:

$$I = I_{ph} - I_0 \left(e^{\frac{q(V+IR_s)}{nkT}} - 1 \right) - \frac{V + IR_s}{R_{sh}}$$

- If $R_s \rightarrow 0$ and $R_{sh} \rightarrow \infty$, the equation simplifies to:

without losses

$$I = I_{ph} - I_0 \left(e^{\frac{qV}{nkT}} - 1 \right)$$

I = output current (A)

V = output voltage (V)

I_{ph} = photocurrent

I_0 = diode reverse saturation current

q = electronic charge (1.602×10^{-19} C)

k = Boltzmann constant (1.381×10^{-23} J/K)

T = cell temperature in Kelvin

n = diode ideality factor (1–2)

R_s = series resistance (Ω)

R_{sh} = shunt resistance (Ω)



Important Parameters

- Short-Circuit Current (I_{sc}): At $V = 0$

$$I_{sc} \approx I_{ph}$$

- Open-Circuit Voltage (V_{oc}): At $I = 0$

$$V_{oc} = \frac{nkT}{q} \ln \left(\frac{I_{ph}}{I_0} + 1 \right)$$

- Maximum Power Point (MPP):

$$P_{mpp} = V_{mpp} \times I_{mpp}$$

- Fill factor (FF):

available area to generate electricity

$$FF = \frac{V_{mpp} \times I_{mpp}}{V_{oc} \times I_{sc}}$$

- Efficiency (η):

$$\eta = \frac{V_{mpp} \times I_{mpp}}{G \times A}$$



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EXAMPLES



Example 1

The ^{higher the better} current density of a solar cell having an area of ^A 100 cm² at STC is given as 35 mA/cm². Find out the output current of the solar cell.

J

$$J = \frac{I}{A}$$

(3.5 A)

$$\begin{aligned} I &= J A \\ &= (35 \times 10^{-3}) (100) \\ &= 3.5 \text{ A} \end{aligned}$$



Example 2

A solar PV module is fabricated using 36 solar cells of 155 cm² connected in series. The cells have 35 mA/cm² current density at STC. Estimate the maximum current produced by the module at STC.

$$I = IA$$

$$I = 35(155) \quad (5.42 \text{ A})$$

$$= 5425 \text{ mA}$$



Example 3

A solar cell gives a current of 0.6 A and voltage of 0.5 V at maximum power point. What is the maximum power point of the solar cell?

$$P = VI$$

$$= 0.5(0.6)$$

$$= 0.3 \text{ W}$$

(0.3 W)



Example 4

A solar cell having an area of 100 cm^2 gives 3.1 A current at maximum power point and 0.5 V at maximum power point at STC. The cell gives 3.5 A short circuit current and 0.6 V open circuit voltage. What is the maximum power point of the solar cell? Also, find out the efficiency of the cell.

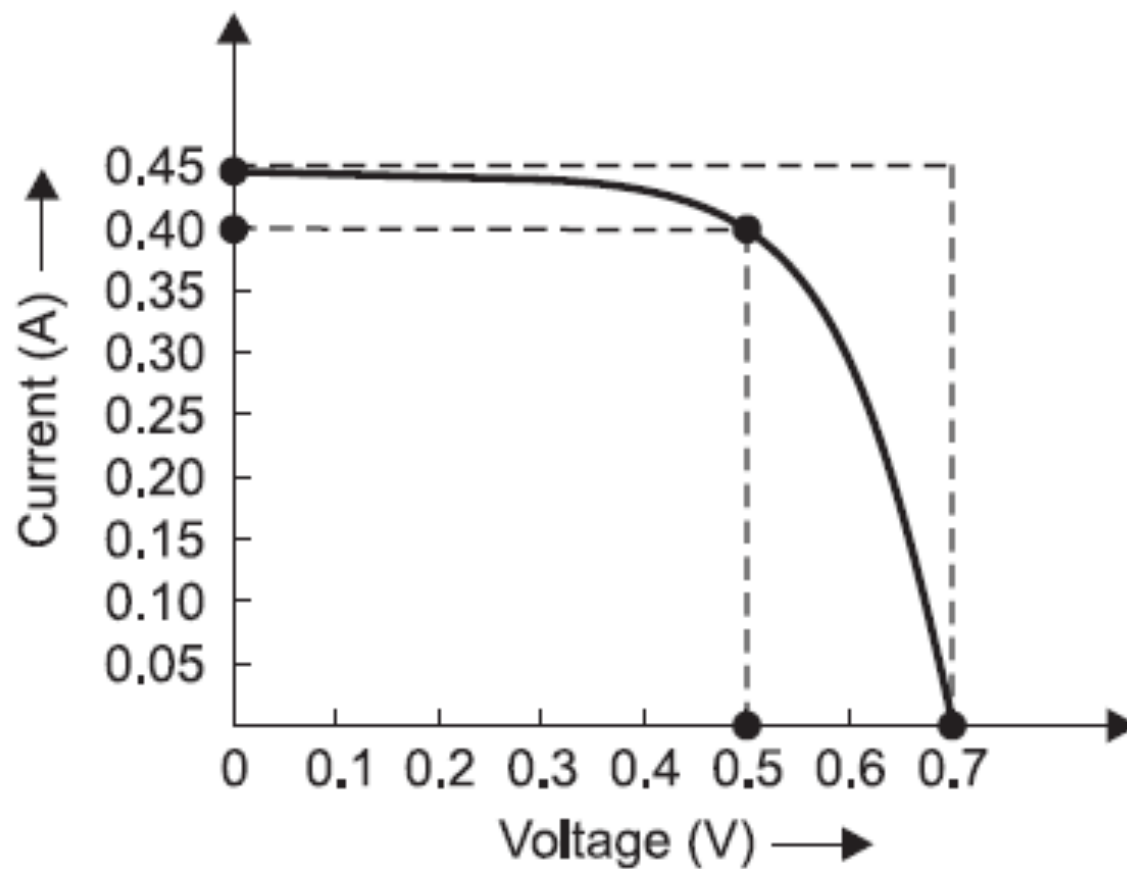
$$\begin{aligned} MPP &= (VI)_{mp} \\ &= 0.5 (3.1) \\ &= 1.55 \text{ W} \end{aligned}$$

$$\begin{aligned} \eta &= \frac{V_{mp} \times I_{mp}}{G A} \\ &= \frac{1.55}{1000 (100 \times 10^{-4})} \\ &= 15.5 \% \end{aligned}$$

(1.55 W, 15.5%)

Example 5

Refer the I-V curve and find out the Fill Factor for the solar cell.



(63.49%)

$$\begin{aligned}\text{Fill Factor} &= \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}} \\ &= \frac{0.5 (0.40)}{0.7 (0.45)} \\ &= 63.49\%\end{aligned}$$



Example 6

A solar cell having an area of 25 cm² gives a current of 0.85 A and voltage 0.55 V at maximum power point. The short circuit current is 0.9 A and open circuit voltage is 0.65 V. What is the Fill Factor, maximum power point and efficiency of the solar cell? Consider STC.

$$M_{pp} = 0.85 (0.55) \\ = 0.4675 \text{ W}$$

$$\text{Fill factor} = \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}}$$

$$= \frac{0.4675}{(0.65)(0.9)}$$

$$= 0.7991$$

$$= 79.91\%$$

$$\eta = \frac{M_{pp}}{G A}$$

$$= \frac{0.4675}{1000 (25 \times 10^{-4})}$$

$$= 0.187$$

$$= 18.7\%$$

(0.4675 W, 79.91%, 18.7%)



Example 7

Calculate the output power from a module if its efficiency is 22%, input power density is 1000 W/m² and area of module is 1.49 m by 0.99 m.

$$\eta = \frac{P_{out}}{G A}$$

(324.5 W)

$$\begin{aligned} P_{out} &= 0.22 (1000) (1.49 \times 0.99) \\ &= 324.52 \text{ W} \end{aligned}$$



Example 8

A solar cell having $FF=60\%$ gives 2.5 A current at maximum power point at STC. The cell gives 3 A short circuit current and 0.5 V open circuit voltage. What is the voltage at maximum power point of the solar cell?

$$FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}$$

(0.36 V)

$$V_{mp} = \frac{FF(V_{oc} I_{sc})}{I_{mp}}$$

$$= \frac{0.6(0.5 \cdot 3)}{2.5}$$

$$= 0.36\text{ V}$$



Example 9

A solar cell having Fill Factor (FF) 68% gives 0.6 V voltage at maximum power point at STC. The cell gives 3 A short circuit current and 0.7 V open circuit voltage. What is the current at maximum power point of the solar cell?

$$I_{mp} = \frac{0.68 (0.7 \cdot 3)}{0.6}$$
$$= 2.38 \text{ A}$$

(2.38 A)



Example 10

Calculate the output power from a solar cell if its efficiency is 30, 24, 19, 16 and 12, input power density is 1000 W/m^2 , and area of the solar cell is 100 cm^2 .

$$\eta = \frac{P_{\text{out}}}{G A}$$

$$P_{\text{out}} = \eta \cdot G A$$

$$= 0.3 (1000) (100 \times 10^{-4})$$

$$= 3 \text{ W}, 2.4 \text{ W}, 1.9 \text{ W}, 1.6 \text{ W}, 1.2 \text{ W}$$

(3W, 2.4W, 1.9W, 1.6W, 1.2W)



Example 11

A Wp rating of a PV module is 230 Wp under STC. What will be the output power of the PV module if the solar radiation intensity is only 400 W/m²?

$$\text{STC, } G = 1000 \text{ W/m}^2$$

(92 Wp)

$$W_p = \frac{400}{1000} (230)$$

$$= 92 \text{ W}_p$$



Example 12

Calculate the output power for solar cells of efficiencies 16%. When the irradiance is say, 1000, 800, 600 and 400 W/m² and area of solar cell is 100 cm².

$$P_{out} = \eta \cdot G A$$

$$= 0.16 (1000) (100 \times 10^{-4})$$

$$(1.6W, 1.28W, 0.96W, 0.64W)$$

$$= 1.6W, 1.28W, 0.96W, 0.64W$$



Example 12

Calculate new value of output current for solar cells of area 20, 30, 50, 80 and 100 cm², when current density of cell is 35 mA/cm².

$$I = 35 \times 20$$

$$= 700 \text{ mA} = 0.7 \text{ A}$$

(0.7A, 1.05A, 1.75A, 2.8A, 3.5A)

$$1.05 \text{ A}, 1.75 \text{ A}, 2.8 \text{ A}, 3.5 \text{ A}$$



Example 13

A solar cell has maximum power point of 0.3 W. The cell voltage at maximum power point at STC is 0.65 V. What is the current at maximum power point of the solar cell?

$$MPP = I_{mp} V_{mp}$$

$$I_{mp} = \frac{0.3}{0.65}$$

$$= 0.46 \text{ A}$$

(0.46 A)



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FACTORS AFFECTING SOLAR CELL POWER

Factors Affecting Solar Cell Power

There are five common factors that affect the power generated by solar cells. They are:

- The conversion efficiency (η),
- The solar irradiation (P_{in}),
- The solar cell area (A)
 - The current output is directly proportional to the cell area.
- The angle at which day light falls (θ)
 - Solar cell produces maximum power when sunlight falls perpendicular to the surface of solar cells. When the light does not perpendicular to solar cells, it always gives less output power.
- The operating temperature (T)
 - When the temperature of solar cell increases; I_{sc} increases while output voltage, efficiency and output power reduces.

Factors Affecting Solar Cell Power

PV Technology name	Typical value of change in parameter value per °C rise in cell temperature from standard test condition (STC) value of 25°C (+ indicates increase, – indicates decrease)			
	Temperature coefficient of current (I_{sc})	Temperature coefficient of voltage (V_{oc})	Temperature coefficient of fill factor (FF)	Temperature coefficient of power (P_m)
Crystalline silicon	+0.08%/°C	–0.35%/°C	–0.15%/°C	–0.45%/°C
Cadmium telluride	+0.04%/°C	–0.25%/°C	–0.035%/°C	–0.25%/°C
Double junction amorphous silicon	+0.07%/°C	–0.3%/°C	–0.095%/°C	–0.25%/°C



Factors Affecting Solar Cell Power

- The change in maximum module voltage due to increase in cell temperature in PV module will be given as:

$$V_{m(temp)} = V_{m(STC)} - TC_{voltage} \times V_{m(STC)} \times \Delta T$$

- The change in module's maximum power output due to increase in cell temperature in PV module will be given as:

$$P_{max(temp)} = P_{max(STC)} - TC_{power} \times P_{max(STC)} \times \Delta T$$

- where TC = temperature coefficient



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EXAMPLES



Example 1

A manufacturer using advanced crystalline silicon cell technology produces the solar cells with V_{oc} of 0.620 V and V_m of 0.510 V at STC. Cell temperature reaches to about 60°C in operating conditions. The modules are designed to connect in grid connected PV plant. If required operating voltage of module is 13.5V, then estimate the number of cells to be connected in series. Given per degree centigrade decrease of voltage for crystalline silicon cell = 0.0023 V.

(32)



Example 2

If the actual operating temperature of the solar cell is 40 °C. The output voltage of a solar cell at standard operating temperature is, say 0.7 V. The output voltage decreases by 2.3 mV/°C. Calculate the new value of output voltage.

<i>Parameter of crystalline silicon solar cells</i>	<i>Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C</i>
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

(0.67V)



Example 3

Efficiency of a crystalline silicon solar cell at STC is 15%. What would be its efficiency when cell operating temperature is 60°C?

<i>Parameter of crystalline silicon solar cells</i>	<i>Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C</i>
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

(12.63%)



Example 4

The maximum rated power point of a crystalline silicon solar cell is 2.5 Wp. Calculate the maximum power output at 45°C.

<i>Parameter of crystalline silicon solar cells</i>	<i>Decrease per °C rise in cell temperature from standard test condition (STC) value of 25°C</i>
Voltage	−0.0023 V or −2.3 mV
Power	−0.45%
Efficiency	−0.45%

(2.275 W)



Example 5

The maximum rated power point of a crystalline silicon solar cell is 75 Wp. Calculate the maximum power output at 55 °C. For crystalline silicon, cell decrease in maximum power per degree centigrade increase in cell temperature is $-0.45\%/^{\circ}\text{C}$.

(64.87 W)



Example 6

Voltage at maximum power point (V_m) of a crystalline silicon solar PV module at STC is 17 V. What would be the voltage when PV module operating temperature is 60 °C? Given temperature coefficient of V_{oc} is $-0.35\%/^{\circ}\text{C}$.

(14.91 V)



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Solar Cell Technology



Common Types of Solar Cells



amorphous silicon



single crystal silicon



polycrystalline silicon



Thin film



Mono silicon



Poly silicon

Common Types of Solar Cells

Type	Materials	Efficiency (%)	Open-circuit voltage (V)	Short-circuit current density (mA/cm ²)	Fill factor (%)
Crystalline Silicon (c-Si) Solar Cell	Mono-crystalline	14 – 16	0.55 – 0.68	30 – 38	70 – 78
	Poly or multicrystalline (mc-Si)	14 – 16	0.55 – 0.68=5	30 – 35	70 – 76
Thin Film Solar Cell	Amorphous Si (a-Si)	6 – 9	0.70 – 1.10	8 – 15	60 – 70
	Cadmium telluride (CdTe)	8 – 11	0.80 – 1.00	15 – 25	60 – 70
	Copper-Indium-Gallium-Selenide (CIGS)	8 – 11	0.50 – 0.70	20 – 30	60 – 70
Multi-Junction Solar Cell	Gallium indium phosphide/ Gallium arsenide/ Germanium (GaInP/GaAs/Ge)	30 – 35	1.00 – 2.50	15 – 35	70 – 85



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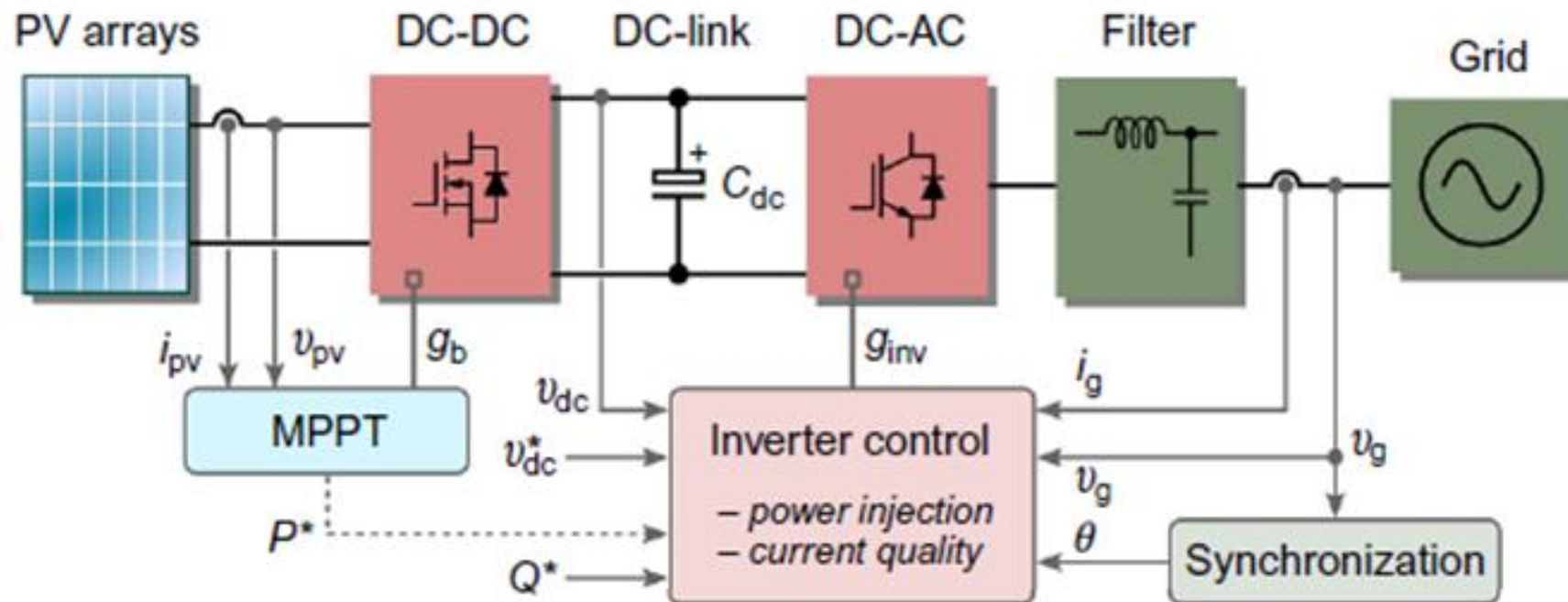
Maximum Power Point Tracking



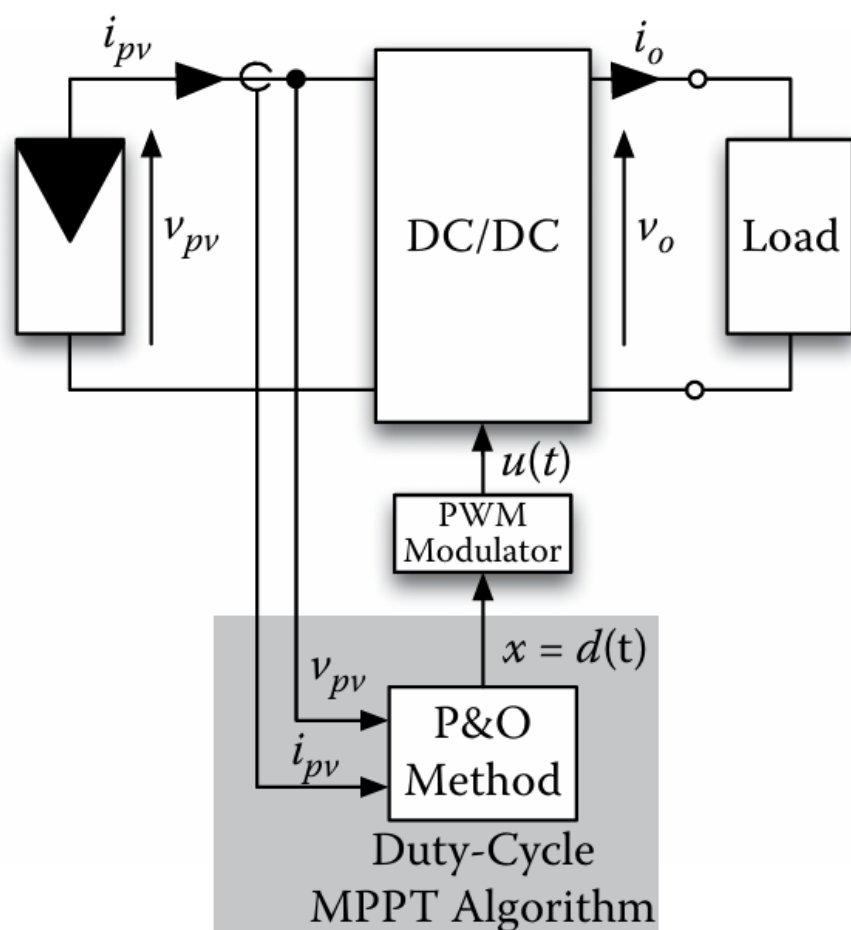
Introduction

- A PV module has a nonlinear I – V characteristic. Power reaches maximum at one point, i.e., the maximum power point (MPP).
- The MPP shifts with irradiance and temperature.
- MPP tracking (MPPT) is a control technique used in PV converters to continuously adjust operating voltage/current so the PV system always extracts the maximum possible power.
- It works by adjusting the operating point of the PV module via a DC-DC converter.
- Perturb and Observe (P&O) and Incremental Conductance (Inc) are the most common MPPT algorithms, while advanced AI-based methods are emerging.

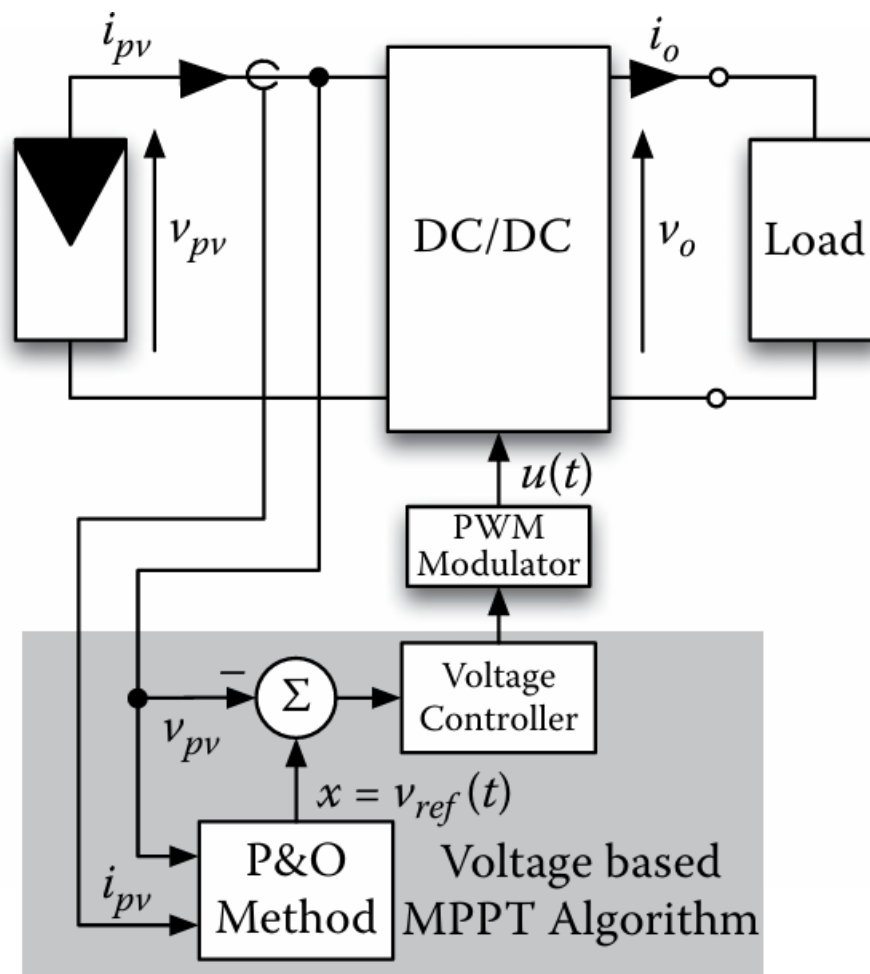
Introduction



Perturb and Observe



P&O without voltage
controller



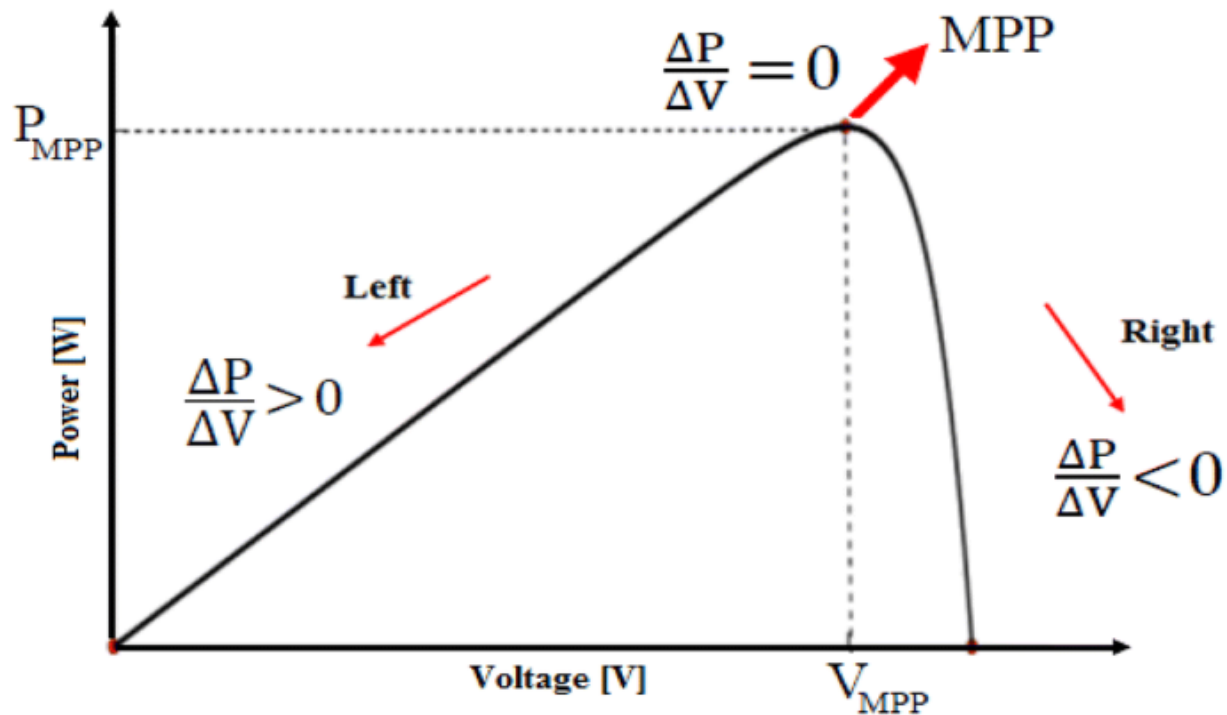
P&O with voltage
controller



Perturb and Observe

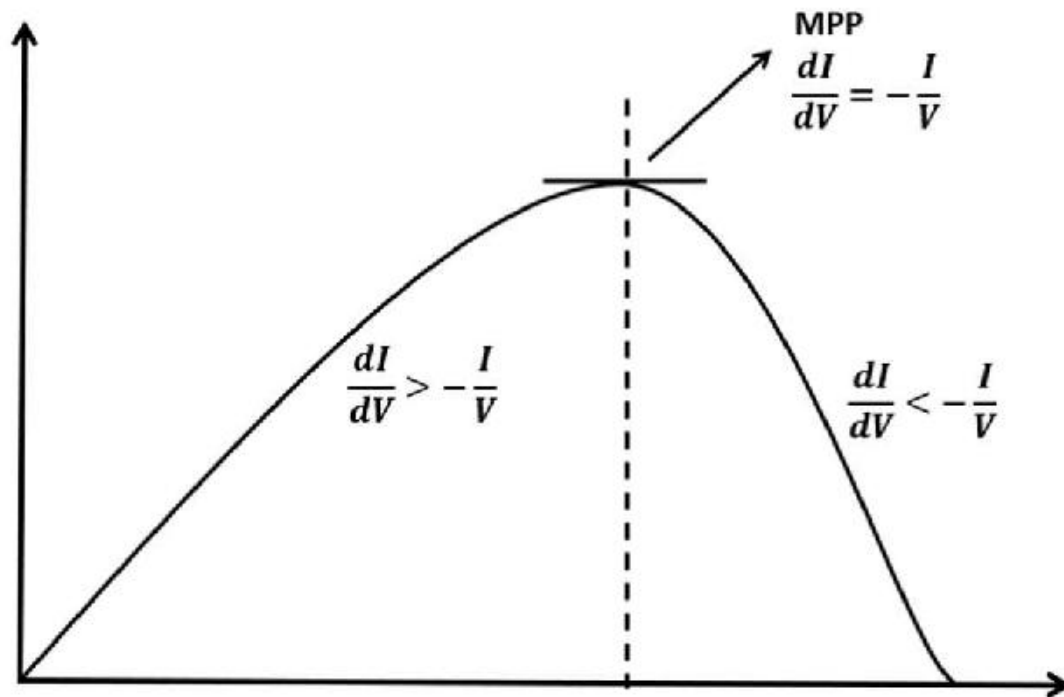
- The P - V curve has a single peak (MPP) under uniform irradiance.
- P&O works by:
 - ☐ Perturbing the operating voltage (duty cycle)
 - ☐ Observing the change in output power
 - ☐ Deciding whether to keep perturbing in the same direction or reverse it
- The perturbation step is selected on the basis of performance trade-off between transient rise time and steady-state conditions.

Perturb and Observe



- If $\frac{\Delta P}{\Delta V} = 0$, MPP is reached
- If $\frac{\Delta P}{\Delta V} > 0$, increase voltage
- If $\frac{\Delta P}{\Delta V} < 0$, decrease voltage

Incremental Conductance



Incremental conductance is measured, i.e.,

$$\frac{dI}{dV}$$

$$\frac{dP}{dV} = \frac{d(IV)}{dV} = I \frac{dV}{dV} + V \frac{dI}{dV} = I + V \frac{dI}{dV}$$

$$\text{At MPP, } \frac{dP}{dV} = 0 \rightarrow \frac{dI}{dV} = -\frac{I}{V}$$



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QUESTIONS?